Last name _____

First name _____

LARSON—MATH 310–CLASSROOM WORKSHEET 21
Matrix Inverses.

Review

1. What is the *inverse* of a (square) matrix A?
2. Find A^{-1}.
3. Use A^{-1} to solve $A\vec{x} = \vec{b}$.
4. Now use the 2×2 inverse formula to find A^{-1}.
5. What is an *orthogonal* matrix Q?
6. We showed $(A^{-1})^T = (A^T)^{-1}$.

Matrix Inverses!

1. Suppose Q is an orthogonal matrix. What is Q invertible? What is Q^{-1}?

2. Why is any lower-triangular matrix L with **non-zero** entries on the diagonal invertible?

3. Why is any upper-triangular matrix U with **non-zero** entries on the diagonal invertible?

4. How can we *find* the inverse of a upper-triangular matrix R with **non-zero** entries on the diagonal?

5. How can we use Gram-Schmidt to find the inverse of a square matrix (with linearly independent columns)?

6. What is *back substitution*? If R is an upper-triangular matrix with non-zero entries on the diagonal, how can we solve any system $R\vec{x} = \vec{b}$?